

Baby Olivia · born at 28 weeks · day 4 of life · feeding poorly, tachypneic · **continuous "machinery" murmur** at the left upper sternal border · widened pulse pressure · echo shows **patent ductus arteriosus**

BIO 004 · WEEK 3 · CASE DEEP DIVE

Vessel Disorders & Fetal Circulation

A premature infant pink one moment and dusky the next. A continuous heart murmur. Three fetal shunts that should have closed but didn't.

PATIENT CASE

Baby Olivia was born at **28 weeks gestation**, weighing 1.2 kg, and has been in the NICU for four days. She did well on the first two days, but today the bedside nurse notes **increasing work of breathing, poor feeding, and pulses that "bound" when palpated**.

The neonatologist auscultates and hears a **continuous murmur** — present in both systole and diastole — at the **left upper sternal border**, often described as "machinery-like." Olivia's **pulse pressure is widened** (BP 65/25).

Echocardiogram confirms a **persistent left-to-right shunt through a patent ductus arteriosus**. The team starts indomethacin.

Before birth, Olivia's circulation looked completely different from yours. Three shunts let her bypass her non-functional lungs and her closed-off placenta. Those shunts should close in the first hours and days of life. In premature infants, they often don't. To understand why Olivia is suddenly in trouble on day four, you have to know how the fetal circulation works and how it transitions at birth.

GUIDING QUESTION

How does blood flow in a fetus, why does that route make sense before birth, and what stops working when a fetal shunt fails to close after the lungs take over?

Three vessel diseases worth knowing

• Aneurysm

Localized dilation of an arterial wall. Most common locations: abdominal aorta (AAA), cerebral arteries (berry aneurysms in the circle of Willis), thoracic aorta. Risk factors: hypertension, smoking, atherosclerosis, connective tissue diseases. Risk: rupture.

• Aortic dissection

Tear in the intima that allows blood to enter and split the layers of the aortic wall. Sudden tearing chest or back pain. Risk factors: hypertension, Marfan syndrome, bicuspid aortic valve. Surgical or medical emergency depending on type.

• Deep vein thrombosis (DVT)

Clot in a deep vein, usually leg. Risk factors (Virchow's triad): stasis (immobility, long flights), hypercoagulability (cancer, pregnancy, OCPs), endothelial injury (surgery, trauma). Risk: pulmonary embolism.

• Varicose veins

Incompetent venous valves let blood pool. Common in the saphenous system of the lower extremity. Multiparity, prolonged standing, and age are risk factors.

Lungs that don't work, a placenta that does

Before birth, the fetus gets oxygen from the placenta, not the lungs. The lungs are collapsed and fluid-filled, with high pulmonary vascular resistance — blood does not want to flow through them. The placenta is highly perfused and where gas exchange happens. The circulation needs to: (1) deliver placental oxygenated blood to the brain and heart, (2) shunt most blood away from the lungs, and (3) return deoxygenated blood to the placenta.

The system solves this with three shunts and an oxygen-rich umbilical vein.

Bypasses that close at birth

• Ductus venosus

Shunt in the fetal liver that carries most umbilical-vein blood directly to the inferior vena cava, bypassing the hepatic circulation. Closes within days of birth → becomes *ligamentum venosum*.

• Foramen ovale

An opening in the interatrial septum that lets oxygenated blood from the right atrium pass directly into the left atrium, bypassing the lungs. Closes functionally at birth as left atrial pressure rises and pushes the flap closed → becomes *fossa ovalis*. Anatomic closure takes weeks to months. Failure = patent foramen ovale (often asymptomatic but can paradoxically embolize).

• Ductus arteriosus

Connects the pulmonary trunk to the aortic arch, shunting blood away from the lungs and into the systemic circulation. Closes within 1-3 days of birth → becomes *ligamentum arteriosum*. Failure to close = patent ductus arteriosus (*Olivia*).

Follow one drop from placenta to placenta

1. Oxygenated blood leaves placenta via the **umbilical vein** (carries oxygen TOWARD the fetus, despite being called a vein).
 2. Most of it bypasses the liver through the **ductus venosus** → IVC → right atrium.
 3. From the right atrium, most flows through the **foramen ovale** directly into the left atrium → left ventricle → aorta → head and upper body (so the brain gets the best-oxygenated blood).
 4. The right atrium's remaining blood (plus SVC return) goes to the right ventricle → pulmonary trunk. But most of that shunts through the **ductus arteriosus** into the aortic arch (after the head vessels), supplying the lower body.
 5. Deoxygenated blood from the fetus returns to the placenta via the paired **umbilical arteries** (despite the name, carry deoxygenated blood AWAY from the fetus).
 6. Gas exchange happens in the placenta. Cycle repeats.
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What changes the moment a baby breathes

Several things happen almost simultaneously when a baby takes its first breath and the cord is clamped:

● Lungs expand

Air fills the alveoli. Pulmonary vascular resistance drops dramatically. Blood flows easily into the lungs for the first time.

● Left atrial pressure rises

More blood returning from now-perfused lungs increases left atrial pressure. The flap of the foramen ovale is pushed closed.

● Placenta disconnected

Cord clamping removes the low-resistance placental circuit. Systemic vascular resistance rises.

● Oxygen tension rises

Oxygenated arterial blood from the now-functioning lungs constricts the ductus arteriosus (oxygen-sensitive smooth muscle), starting its closure.

● Prostaglandins fall

Placental prostaglandins, which kept the ductus open in utero, disappear after birth. This combines with rising oxygen to close the ductus.

What goes wrong in prematurity. The ductus arteriosus in premature infants is less responsive to oxygen and more dependent on prostaglandin levels. It often fails to close after birth. Once the lungs take over, blood now flows from the high-pressure aorta into the low-pressure pulmonary artery (left-to-right shunt), flooding the lungs and stressing the left heart.

Why Olivia is in trouble

With a persistent ductus arteriosus, blood that should flow from the aorta to the body is short-circuited into the pulmonary trunk. Consequences:

- **Pulmonary overflow** from increased pulmonary blood flow → pulmonary edema, increased work of breathing.
- **Systemic underflow** distally — particularly to the gut and kidneys — risk of necrotizing enterocolitis and acute kidney injury.
- **Volume overload of the left heart**, which must accept extra returning blood.
- **Wide pulse pressure** (high systolic, low diastolic) because blood is "running off" through the PDA during diastole, dropping diastolic BP.
- **Continuous "machinery" murmur**: aorta's pressure exceeds pulmonary trunk's throughout the cardiac cycle, so blood flows through the duct in both systole and diastole.



DRAW ZONE 1 -- Diagram fetal circulation. Show umbilical vein, ductus venosus, foramen ovale, ductus arteriosus, umbilical arteries. Use color coding for oxygenation.

Star each shunt and label what it becomes after birth (ligamentum venosum, fossa ovalis, ligamentum arteriosum).



DRAW ZONE 2 -- Draw the PDA. Show the aortic arch and pulmonary trunk with the persistent connection. Use arrows for the left-to-right shunt and show pulmonary overflow.

Mark the location of the continuous murmur (left upper sternal border) and label the wide pulse pressure pathophysiology.

RETURNING TO OLIVIA

Three shunts, two of them closed, one that didn't

Olivia's circulation is a textbook example of why timing of fetal shunt closure matters. Her ductus arteriosus was useful in utero. Open after birth, it's drowning her lungs.

"Continuous machinery murmur, LUSB"

Blood flows from aorta to pulmonary artery in BOTH systole and diastole through the patent duct. Aortic pressure exceeds pulmonary pressure throughout the cardiac cycle. The murmur sounds like continuous machinery.

"Widened pulse pressure"

During diastole, blood escapes from aorta through the PDA → drops aortic diastolic pressure. Systolic stays high because LV is volume-overloaded. Wide pulse pressure ($65 - 25 = 40$ with low diastolic) and bounding pulses are classic.

"Increased work of breathing day 4"

Pulmonary overflow + pulmonary edema from the left-to-right shunt. The lungs are flooded with blood from the aortic shunt; gas exchange suffers.

"Poor feeding"

Increased work of breathing makes coordinated suck-swallow-breathe difficult. Also gut hypoperfusion (blood diverted into lungs instead of to gut) reduces tolerance of feeds.

"Why indomethacin"

Indomethacin (and ibuprofen) inhibit prostaglandin synthesis. Lower prostaglandin levels promote ductal closure. About 70% of preterm PDAs close with medical therapy.

"If indomethacin fails"

Surgical ligation or transcatheter device closure. Larger PDAs in older infants are increasingly closed with a catheter-delivered occluder.

Shunt, transition, prostaglandin, closure. Olivia's case shows the fetal circulation in negative — what happens when one piece doesn't make the transition. When you go back to the prework, the fetal-to-newborn flow diagram will land differently — Olivia's duct is exactly the one that should have closed by now.

